

# The separation complexity of hypermetric and binary quadratic inequalities

## Abstract

We prove that exact separation of hypermetric inequalities is strongly NP-complete. The proof gives a rational Gram construction in which the violated inequalities correspond exactly to exact covers. A scaling with polynomially bounded denominator places the hard points in the metric polytope and the interior of the ellipsope. For any fixed gonality cutoff, the construction can also satisfy every rounded positive semidefinite inequality below that cutoff. The same construction, together with switching and an orthogonal extension, proves strong NP-completeness for gap-1, rounded positive semidefinite, Boros–Hammer, Boolean quadric clique and cut, pure hypermetric, and odd clique separation. Hardness for odd cliques already holds for the unswitched all-positive inequalities. The constructed clique, cut, pure hypermetric, and odd clique inequalities define facets; we give a direct proof. We also give polynomial-size certificates, an exact rank-parameterized lattice algorithm, and an explicit algorithm for violations above a normalized threshold. Finally, we prove NP-completeness of gap-0 separation at metric-feasible points outside the ellipsope; this reduction does not establish strong hardness. These results distinguish the difficulty of unrestricted exact separation from the tractability of semidefiniteness tests, fixed-rank inputs, and restricted searches.

**Keywords:** separation; hypermetric inequalities; Boolean quadric polytope; quadratic integer programming; semidefinite relaxation.

## 1 Introduction

Hypermetric inequalities connect cut polytopes, quadratic optimization, and the geometry of lattices. They are valid for every cut, and their Boolean quadric images strengthen the usual linear and semidefinite relaxations of binary quadratic problems. We study the exact decision problem of whether a rational point violates any hypermetric inequality. We prove that this problem is strongly NP-complete. The hardness persists at points satisfying all triangle and perimeter inequalities, whose ellipsope matrix is positive definite, and which satisfy every rounded positive semidefinite inequality up to any prescribed fixed gonality cutoff.

The proof also determines the separation complexity of several related families. These include rounded positive semidefinite and gap-1 inequalities in cut form, Boros–Hammer inequalities in Boolean quadric form, the Boolean quadric clique and cut inequalities, pure hypermetric inequalities, and odd clique inequalities. For the last family, hardness holds even when only the unswitched inequalities with all nonzero coefficients equal to 1 are available. These conclusions do not follow merely from inclusion among inequality families. We construct points for which every possible violation in the larger families encodes the same exact cover, and use switching and a small orthogonal extension to obtain the families with coefficients in  $\{0, \pm 1\}$ .

## 1.1 Relation to prior work

Avis and Grishukhin [2] formulated the recognition problem for the hypermetric cone, established its membership in coNP, and related it to finding lattice points inside a sphere. They also proved hardness for problems in which a bound on the gonality of the inequality is part of the input. Avis [1] extended this complexity analysis to negative-type,  $k$ -gonal, and gap inequalities. These results leave a logical distinction: restricting the coefficient norm can be hard even when separation over the full class is easy. Negative-type inequalities illustrate the distinction, because full separation is a positive semidefiniteness test. The results for a prescribed gonality therefore do not establish hardness of unrestricted hypermetric separation. Deza and Grishukhin [8] also treated the equivalent hypermetric correlation recognition problem and described its NP-hardness as unknown.

The Boolean quadric families studied here have a separate polyhedral history. Padberg [21] developed the clique and cut inequalities, and Boros and Hammer [4] studied the more general integer-coefficient quadratic inequalities now bearing their names. The covariance map relates these families to cut inequalities; in particular, the Boros–Hammer class is the Boolean quadric form of the rounded positive semidefinite class. Letchford and Sørensen [19] established separation reductions among these classes and hypermetric correlation inequalities. Galli et al. [10, 11] placed these families within the theory of gap inequalities, originating in Laurent and Poljak [17], and described the remaining complexity questions. Letchford [18] again identified hypermetric separation as an open problem and recorded unknown separation complexity for the clique, cut, and related Boolean quadric families. The NP-completeness of testing  $\gamma(b) = 0$  for a specified vector  $b$  was already known [11, §2]. It does not determine gap-0 separation, which asks whether a given point has any violating coefficient vector with zero gap.

Two secondary accounts require a qualification of this history. Liers [20, p. 23] states that hypermetric separation is NP-hard and cites Chapter 28 of Deza and Laurent [9]. Krishnan and Terlaky [16, §6, “The hypermetric inequalities”] make an analogous assertion for the gap-1 class, which they call hypermetric, citing Avis [1], Avis and Grishukhin [2]. Neither discussion supplies a reduction. The primary complexity papers and the later survey explicitly leave the unrestricted question open. We therefore distinguish those assertions from a complete proof of the exact problem and its relaxation promises.

Hardness of separating split cuts in mixed-integer linear programming [6] does not settle these questions. Here the input is a quadratic moment matrix, and in the binary case its diagonal entries must equal its first-row entries. At positive definite moment points, these identities place the origin and the lattice generators on a common sphere in the hypermetric representation. A reduction for an unrestricted integer quadratic form need not preserve them. The construction below uses the fact that every 0–1 vector lies on the sphere through the origin centered at  $\mathbf{1}/2$ . It couples this sphere with exact-cover incidence vectors and controls all integer points, including coefficient vectors of arbitrarily large norm.

Conversely, the binary hard points are admissible general integer moment matrices. The same reduction therefore proves strong NP-completeness of unrestricted integer-quadratic split separation, whose complexity was left open by Buchheim and Traversi [5]. This consequence requires no separate integer construction.

Recent QCQP work also searches for Boros–Hammer inequalities by solving nonconvex integer quadratic subproblems with variable coefficients restricted to  $[-2, 2]$ , in computational evaluations of semidefinite relaxations [7, §4.1.2].

For algorithmic context, integer-quadratic separation methods can exploit low rank and lattice structure [5]. Our algorithmic results give explicit reductions to exact closest-vector problems in rational lattices. Existing closest-vector algorithms then yield bounds in the binary input model.

We distinguish rank from support because they lead to different parameter dependence. We do not claim a new closest-vector algorithm.

## 1.2 Contributions and their scope

The main contributions are as follows.

1. We give an exact-cover Gram construction whose violated hypermetric inequalities correspond exactly to exact covers ([Theorem 3.1](#)). Its negative slacks are all  $-1/2$  in the correlation representation. A polynomially encoded scaling places the distance point in  $\text{MET}(V)$  and its matrix  $Z$  in the interior of  $\text{ELL}(V)$ , while excluding every rounded positive semidefinite violation with coefficient sum of absolute value at least 3 ([Lemma 3.2](#) and [Theorem 3.3](#)). This proves strong NP-completeness of unrestricted hypermetric separation ([Corollary 3.5](#)).
2. We transfer this construction to the gap-1, rounded positive semidefinite, Boros–Hammer, and signed-coefficient Boolean quadric families ([Theorems 4.1](#) and [4.3](#)). A further orthogonal extension proves strong NP-completeness for pure hypermetric and odd clique separation, including unswitched positive odd clique inequalities ([Theorem 4.4](#)). The witnesses for the clique, cut, pure hypermetric, and odd clique families define facets; we include a direct proof ([Lemma 4.2](#)).
3. We provide polynomial-size certificates for the infinite families, an exact algorithm parameterized by rank, and an explicit enumeration bound for separation above a normalized violation threshold ([Section 5](#)). The threshold bound is an XP bound: its exponent depends on the reciprocal threshold. A separate general-integer example shows why restricting the allowed coefficient vectors can destroy the tractability of low-rank separation.
4. We prove NP-completeness of gap-0 separation, with hard points already in the metric polytope ([Theorem 6.2](#)). These points are outside the ellipsope. Gap-0 inequalities cannot cut a positive semidefinite point, so this result has a different algorithmic role from the main theorem. Its reduction uses PARTITION; we make no strong-hardness claim for this family.

To the best of our knowledge, no prior work gives complete proofs of the unrestricted hardness results in the first, second, and fourth items with the stated domains and promises. The novelty is the exact decision complexity of the stated families and the explicit relaxation promises, rather than their definitions, the covariance and switching operations, or the use of closest-vector methods. The earlier open-problem statements and separation equivalences support this qualified claim; the secondary assertions described above are part of its historical context.

These are exact separation results. They rule out polynomial-time exact oracles for the hard families unless  $P = NP$ , even after the standard relaxations and any fixed gonality cutoff have been enforced. They do not compare cut-selection heuristics or establish hardness at a fixed additive tolerance. The constructed violations have inverse polynomial size, so hardness is not caused by exponentially small numerical slacks. Their Euclidean-normalized size can nevertheless decrease with the instance dimension; this is consistent with the threshold algorithm in [Section 5](#).

The paper first fixes the representations and separation conventions ([Section 2](#)). [Section 3](#) develops the exact-cover construction and its scaling. [Section 4](#) treats the related binary families. [Section 5](#) gives certificates and algorithms, and [Section 6](#) treats gap-0 separation. [Section 7](#) discusses the implications and the remaining question for general gap inequalities at positive semidefinite points.

Table 1: Cut inequality families. The last row defines one class for each positive integer  $k$ .

| Family                        | Coefficients                                  | Inequality                                   |
|-------------------------------|-----------------------------------------------|----------------------------------------------|
| Hypermetric                   | $b \in \mathbb{Z}^V, \sigma(b) = 1$           | $Q(b, d) \leq 0$                             |
| Negative type                 | $b \in \mathbb{Z}^V, \sigma(b) = 0$           | $Q(b, d) \leq 0$                             |
| Positive semidefinite         | $b \in \mathbb{R}^V$                          | $Q(b, d) \leq \sigma(b)^2/4$                 |
| Rounded positive semidefinite | $b \in \mathbb{Z}^V, \sigma(b) \text{ odd}$   | $Q(b, d) \leq (\sigma(b)^2 - 1)/4$           |
| Gap                           | $b \in \mathbb{Z}^V$                          | $Q(b, d) \leq (\sigma(b)^2 - \gamma(b)^2)/4$ |
| Gap-1, gap-0                  | $\gamma(b) = 1, \gamma(b) = 0$                | As in the gap row                            |
| Odd clique                    | $b \in \{0, \pm 1\}^V, \sigma(b) \text{ odd}$ | $Q(b, d) \leq (\sigma(b)^2 - 1)/4$           |
| Pure hypermetric              | $b \in \{0, \pm 1\}^V, \sigma(b) = 1$         | $Q(b, d) \leq 0$                             |
| $k$ -gonal                    | $b \in \mathbb{Z}^V, \sum_i  b_i  = k$        | $Q(b, d) \leq \lfloor \sigma(b)^2/4 \rfloor$ |

## 2 Inequalities, representations, and separation

### 2.1 Cut inequalities

Let  $V = \{0\} \cup W$  be a finite vertex set with a distinguished root 0. Write  $E(V)$  for the unordered pairs of distinct vertices. A cut vector  $d^S \in \{0, 1\}^{E(V)}$ , for  $S \subseteq V$ , has  $d_{ij}^S = 1$  exactly when one of  $i, j$  belongs to  $S$ . The cut polytope is  $\text{CUT}(V) = \text{conv}\{d^S : S \subseteq V\}$ . For  $d \in \mathbb{R}^{E(V)}$  and  $b \in \mathbb{Z}^V$ , put

$$\sigma(b) = \sum_{i \in V} b_i, \quad Q(b, d) = \sum_{\{i, j\} \in E(V)} b_i b_j d_{ij}, \quad \gamma(b) = \min_{s \in \{\pm 1\}^V} |s^\top b|. \quad (1)$$

We call  $\gamma(b)$  the gap of  $b$ . It has the same parity as  $\sigma(b)$ , and  $0 \leq \gamma(b) \leq |\sigma(b)|$ .

Table 1 fixes the names used in this paper. The positive semidefinite (psd) inequalities allow real coefficients; the other rows use integer coefficients. These conventions follow the comparison of the families in Galli et al. [11]. In particular, a rounded positive semidefinite inequality need not be hypermetric: its coefficient sum can be any odd integer. The  $k$ -gonal class in the table fixes the  $\ell_1$  norm of the coefficient vector, rather than its sum. We call  $\sum_i |b_i|$  its gonality.

Every inequality in the table is valid for  $\text{CUT}(V)$ . Indeed, on a cut  $S$  let  $a = \sum_{i \in S} b_i$ . Then  $Q(b, d^S) = a(\sigma(b) - a) = (\sigma(b)^2 - (2a - \sigma(b))^2)/4$ . The gap bound follows by minimizing the absolute signed sum. The other bounds follow from this identity, parity, or the nonnegativity of a square. The same argument for real  $b$  proves the positive semidefinite bound. Hypermetric coefficients have gap 1, so

$$\text{hypermetric} \subseteq \text{gap-1} \subseteq \text{rounded positive semidefinite}. \quad (2)$$

Containment alone does not transfer hardness of separation: a larger family may cut both the yes and the no instances of a reduction.

Let  $D$  be the symmetric matrix with zero diagonal and off-diagonal entries  $d_{ij}$ , and let  $J = \mathbf{1}\mathbf{1}^\top$ . Set

$$Z(d) = J - 2D. \quad (3)$$

Then  $b^\top Z(d)b = \sigma(b)^2 - 4Q(b, d)$ . The ellipsope  $\text{ELL}(V) = \{Z \succeq 0 : \text{diag } Z = \mathbf{1}\}$  therefore enforces every positive semidefinite inequality. Its relative interior consists of its positive definite matrices. We also use the metric polytope  $\text{MET}(V)$ , defined by the triangle inequalities  $d_{ij} \leq d_{ik} + d_{jk}$  and perimeter inequalities  $d_{ij} + d_{ik} + d_{jk} \leq 2$  for distinct  $i, j, k$ . For  $|V| \geq 3$  these imply  $0 \leq d_{ij} \leq 1$ .

## 2.2 Correlation and moment representations

The following linear bijection sends a distance vector to a symmetric matrix on  $W$ :

$$M(d)_{ii} = d_{0i}, \quad M(d)_{ij} = \frac{d_{0i} + d_{0j} - d_{ij}}{2} \quad (i \neq j). \quad (4)$$

Its inverse is  $d_{0i} = M_{ii}$  and  $d_{ij} = M_{ii} + M_{jj} - 2M_{ij}$ . Write  $\delta = \text{diag } M$  and

$$g_M(z) = z^\top M z - \delta^\top z. \quad (5)$$

For every  $b = (b_0, z) \in \mathbb{R}^V$ , direct substitution gives

$$Q(b, d) = \sigma(b) \delta^\top z - z^\top M z. \quad (6)$$

In particular,  $z \mapsto (1 - \mathbf{1}^\top z, z)$  is a bijection from  $\mathbb{Z}^W$  to the hypermetric coefficient vectors, and

$$Q((1 - \mathbf{1}^\top z, z), d) = -g_M(z). \quad (7)$$

Thus hypermetric separation asks whether an integer  $z$  satisfies  $g_M(z) < 0$ . The inequalities  $g_M(z) \geq 0$ ,  $z \in \mathbb{Z}^W$ , are called hypermetric correlation inequalities. Negative-type membership is equivalent to  $M \succeq 0$ , so a violation exists exactly when  $M$  has a negative direction: the vectors with coefficient sum zero give  $Q(b, d) = -z^\top M z$ . Although the negative-type row of [Table 1](#) uses integers, a rational matrix with a negative real direction also has a rational, and hence an integer, negative direction.

If  $M$  is the Gram matrix of vectors  $u_i$  and there is a point  $c$  with  $2\langle c, u_i \rangle = \|u_i\|^2$  for each  $i$ , then

$$g_M(z) = \left\| \sum_i z_i u_i - c \right\|^2 - \|c\|^2. \quad (8)$$

The points  $0, u_i$  lie on the sphere with center  $c$  through  $0$ . A violated hypermetric inequality is an integer combination strictly inside this sphere. For rational  $M$ , these combinations form a lattice in their span, as follows from [Lemma 5.2](#). For positive definite  $M$ , the center in the span of the  $u_i$  always exists. This geometric description underlies the earlier lattice approach to hypermetricity [\[2\]](#); our reduction specifies the sphere and all of its interior lattice points.

For the Boolean quadric representation, use variables  $x_i$  and  $y_{ij}$ ,  $i, j \in W$ ,  $i < j$ , and put

$$Y(x, B) = \begin{pmatrix} 1 & x^\top \\ x & B \end{pmatrix}, \quad B_{ii} = x_i, \quad B_{ij} = y_{ij} \quad (i \neq j). \quad (9)$$

The Boolean quadric polytope  $\text{BQP}(W)$  is the convex hull of  $(x, y) = (z, (z_i z_j)_{i < j})$  for  $z \in \{0, 1\}^W$ . The covariance map

$$d_{0i} = x_i, \quad d_{ij} = x_i + x_j - 2y_{ij} \quad (10)$$

is an affine bijection between  $\text{BQP}(W)$  and  $\text{CUT}(V)$  [\[18, 21\]](#). For arbitrary points in these ambient spaces,  $B = M(d)$  and  $x = \text{diag } B$ . Moreover,

$$Z(d) = LY(x, B)L^\top, \quad L = \begin{pmatrix} 1 & 0 \\ \mathbf{1} & -2I \end{pmatrix}. \quad (11)$$

Consequently  $Y \succ 0$  if and only if  $Z(d) \succ 0$ .

Some algorithms below apply more generally to a rational moment matrix  $Y$  with  $Y_{00} = 1$ , without the identities  $B_{ii} = x_i$ . We call this the general integer moment setting. For  $v \in \mathbb{Z}^{\{0\} \cup W}$  define the split slack

$$q_Y(v) = v^\top Y v + e_0^\top Y v. \quad (12)$$

It is the linearization of  $(v_0 + v_W^\top z)(v_0 + v_W^\top z + 1)$ , which is nonnegative for every integer  $z$ . In Boolean data these are precisely the Boros–Hammer inequalities [4, 19]. For  $w \in \mathbb{Z}^W$  and  $s \in \mathbb{Z}$ , it is convenient to write their slack as

$$h_Y(w, s) = w^\top B w - (2s + 1)x^\top w + s(s + 1) = q_Y((s, -w)) = q_Y((-s - 1, w)). \quad (13)$$

The two data pairs  $(w, s)$  and  $(-w, -s - 1)$  define the same polynomial. With

$$\beta(w, s) = (2s + 1 - \mathbf{1}^\top w, w) \quad (14)$$

and Boolean data, Equation (6) gives

$$h_Y(w, s) = \frac{\sigma(\beta(w, s))^2 - 1}{4} - Q(\beta(w, s), d). \quad (15)$$

This is a bijective correspondence between Boros–Hammer data and rounded positive semidefinite coefficients. Equivalently, for  $u = e_0 + 2v$  one has

$$q_Y(v) = \frac{u^\top Y u - 1}{4}. \quad (16)$$

For later use, the signed-coefficient Boros–Hammer family takes disjoint subsets  $S, T \subseteq W$  and  $w = \mathbf{1}_S - \mathbf{1}_T$ . Dividing Equation (13) by 2 yields

$$s \sum_{i \in S} x_i + \sum_{i \in S, j \in T} y_{ij} \leq (s + 1) \sum_{i \in T} x_i + \sum_{\{i, j\} \subseteq S} y_{ij} + \sum_{\{i, j\} \subseteq T} y_{ij} + \frac{s(s + 1)}{2}. \quad (17)$$

We call the  $T = \emptyset$ ,  $0 \leq s \leq |S| - 1$  subfamily the clique inequalities, and the  $s = 0$  subfamily the cut inequalities. These are the established Boolean quadric families surveyed by Letchford [18]. Their violations in this divided form are half the corresponding cut-form violations in Equation (15).

## 2.3 Switching and the decision problem

Switching a distance vector on  $U \subseteq V$  replaces  $d_{ij}$  by  $1 - d_{ij}$  when exactly one endpoint lies in  $U$ , and leaves it unchanged otherwise. Write this vector as  $d^U$  and set  $\varepsilon_i = -1$  for  $i \in U$ ,  $\varepsilon_i = 1$  otherwise. Then  $Z(d^U) = \text{diag}(\varepsilon)Z(d)\text{diag}(\varepsilon)$ . Switching preserves the cut and metric polytopes and positive definiteness of  $Z$ . It sends a rounded positive semidefinite inequality with vector  $b$  to one with vector  $\text{diag}(\varepsilon)b$ : parity is unchanged. It also preserves the gap and the restrictions  $b_i \in \{0, \pm 1\}$ . It does not in general preserve the condition  $\sigma(b) = 1$  defining the hypermetric class.

All complexity statements concern rational input, encoded in binary. Let  $L$  denote its total bit length. Exact separation is the decision problem of whether at least one inequality in the specified family is strictly violated. A separation algorithm must also return such an inequality on a yes instance. There is no numerical tolerance in this definition. The complementary membership problem asks whether every inequality of the family is satisfied.

For a rational-input problem, strong NP-hardness here means hardness already on reductions in which every numerator and denominator is bounded by a polynomial in the combinatorial size of the

source instance. The bounds below are stronger: each reduction point has a common denominator bounded by such a polynomial. For infinite families, membership in NP also requires a violated inequality with polynomial bit length; [Section 5](#) proves the required bounds. Promise statements such as  $d \in \text{MET}(V)$  and  $Z(d) \succ 0$  specify properties of all constructed inputs, independently of whether they are yes or no instances.

A running time  $f(r) \text{poly}(L)$  is fixed-parameter tractable in  $r$ : the exponent of  $L$  is independent of  $r$ . An XP bound permits that exponent to depend on the parameter and therefore gives polynomial time for each fixed parameter value. We distinguish these two guarantees throughout.

### 3 An exact-cover reduction

Let  $S_1, \dots, S_n \subseteq [p]$ , where  $p \geq 2$ , and let  $A \in \{0, 1\}^{p \times n}$  be their incidence matrix. An exact cover is a vector  $x \in \{0, 1\}^n$  satisfying  $Ax = \mathbf{1}_p$ . Set  $W = \{1, \dots, n, g\}$ , where  $g$  is an additional index. Choose a rational number  $\eta^2$  such that

$$\max \left\{ \frac{p}{12}, \frac{p-2}{4} \right\} \leq \eta^2 < \frac{p}{4}, \quad (18)$$

and define the symmetric matrix  $M$  on  $W$  by

$$M_{ij} = 4\mathbf{1}_{i=j} + |S_i \cap S_j|, \quad M_{ig} = \frac{|S_i|}{2}, \quad M_{gg} = \frac{p}{4} + \eta^2. \quad (19)$$

The interval in (18) is nonempty for every  $p \geq 2$ ; the choice  $\eta^2 = (p-1)/4$  belongs to it.

**Theorem 3.1** (Exact-cover encoding). *The matrix  $M$  in (19) is positive definite. For every integer vector  $z$ , the following equivalence holds:*

$$g_M(z) < 0 \iff z = (x, -1), \quad x \in \{0, 1\}^n, \quad Ax = \mathbf{1}_p. \quad (20)$$

Every such vector has

$$g_M(z) = -\left(\frac{p}{2} - 2\eta^2\right) \in [-1, 0).$$

*Proof.* Take  $\eta = \sqrt{\eta^2} > 0$ . In  $\mathbb{R}^{n+p+1}$ , define

$$u_i = (2e_i, A_i, 0), \quad u_g = (\mathbf{0}_n, \frac{1}{2}\mathbf{1}_p, \eta),$$

where  $A_i$  is column  $i$  of  $A$ . Their Gram matrix is  $M$ . They are linearly independent: the first block forces every coefficient of the  $u_i$  to vanish, and the last coordinate then forces the coefficient of  $u_g$  to vanish. Thus  $M \succ 0$ .

For  $z = (x, k) \in \mathbb{Z}^n \times \mathbb{Z}$  and  $y = Ax$ , expansion gives

$$g_M(x, k) = 4 \sum_i x_i(x_i - 1) + \sum_e (y_e^2 + (k-1)y_e) + k(k-1) \left( \frac{p}{4} + \eta^2 \right). \quad (21)$$

Indeed, the quadratic part is

$$4 \|x\|^2 + \left\| y + \frac{k}{2} \mathbf{1}_p \right\|^2 + k^2 \eta^2,$$

and the linear part is  $\sum_i (4 + |S_i|)x_i + k(p/4 + \eta^2)$ . The equality  $\sum_i |S_i| x_i = \sum_e y_e$  follows from the incidence matrix.

Every term  $x_i(x_i - 1)$  is nonnegative on the integers. For  $k = 0$ , the element terms in (21) are  $y_e(y_e - 1) \geq 0$ ; for  $k = 1$ , they are  $y_e^2 \geq 0$ . For  $k \geq 2$ , their real minima give

$$g_M(x, k) \geq -\frac{p(k-1)^2}{4} + k(k-1) \left( \frac{p}{4} + \eta^2 \right) = (k-1) \left( \frac{p}{4} + k\eta^2 \right) > 0.$$

Next let  $k = -t$  with  $t \geq 2$ . The minimum of  $y^2 - (t+1)y$  over integer  $y$  is  $-\lfloor (t+1)^2/4 \rfloor$ . If  $t$  is odd, then  $t \geq 3$  and

$$g_M(x, -t) \geq -\frac{p(t+1)^2}{4} + t(t+1) \left( \frac{p}{4} + \eta^2 \right) = (t+1) \left( t\eta^2 - \frac{p}{4} \right) \geq 0.$$

If  $t$  is even, then

$$g_M(x, -t) \geq -\frac{pt(t+2)}{4} + t(t+1) \left( \frac{p}{4} + \eta^2 \right) = t \left( (t+1)\eta^2 - \frac{p}{4} \right) \geq 0.$$

Both last inequalities follow from  $3\eta^2 \geq p/4$ .

Finally, at  $k = -1$ , (21) becomes

$$g_M(x, -1) = 4 \sum_i x_i(x_i - 1) + \|Ax - \mathbf{1}_p\|^2 - \left( \frac{p}{2} - 2\eta^2 \right). \quad (22)$$

The last parenthesis belongs to  $(0, 1]$  by (18). The other two summands are nonnegative integers. Their sum is strictly smaller than that parenthesis exactly when both vanish, which is equivalent to  $x \in \{0, 1\}^n$  and  $Ax = \mathbf{1}_p$ . This proves both directions of (20) and the claimed slack.  $\square$

The construction uses the incidence identity  $\|A_i\|^2 = \mathbf{1}_p^\top A_i$ . Geometrically, every incidence column lies on the sphere through the origin centered at  $\mathbf{1}_p/2$ . The first block of the generators penalizes integer coordinates outside  $\{0, 1\}$ , and the distinguished generator selects the exact-cover equations in the layer  $k = -1$ . The realization may use an irrational  $\eta$ ; only the rational entries of  $M$  are part of the output.

For the rest of this section, fix  $\eta^2 = (p-1)/4$ . The distance vector  $d = d(M)$  is then

$$\begin{aligned} d_{0i} &= 4 + |S_i|, & d_{0g} &= \frac{2p-1}{4}, \\ d_{ij} &= 8 + |S_i \triangle S_j| \quad (i \neq j), & d_{ig} &= \frac{2p+15}{4}. \end{aligned} \quad (23)$$

In particular,  $4d$  is integral with positive entries at most  $4p+32$ . The identity (7) and Theorem 3.1 show that the complete set of hypermetric violators is

$$b_T = (2 - |T|)e_0 + \sum_{i \in T} e_i - e_g, \quad T \text{ an exact cover}, \quad (24)$$

and each satisfies  $Q(b_T, d) = 1/2$ .

To place the reduction inside the standard relaxations, write

$$s = \delta^\top M^{-1} \delta, \quad N = 8n + 2p + 1, \quad \varepsilon = \frac{1}{N}, \quad (25)$$

and define the binary moment matrix

$$Y_\varepsilon = \begin{pmatrix} 1 & \varepsilon \delta^\top \\ \varepsilon \delta & \varepsilon M \end{pmatrix}. \quad (26)$$



**Lemma 3.2** (Polynomial scaling). *For every exact-cover instance with  $p \geq 2$ ,*

$$\max_{i \in W} M_{ii} \leq s \leq 4n + p + \frac{1}{4(p-1)} < \frac{N}{2}. \quad (27)$$

The matrix  $Y_\varepsilon$  satisfies  $(Y_\varepsilon)_{ii} = (Y_\varepsilon)_{0i}$  and

$$Y_\varepsilon - \theta e_0 e_0^\top \succ 0 \quad \text{for every real } \theta \leq \frac{1}{2}. \quad (28)$$

All entries of  $4NY_\varepsilon$  are integers in  $[0, 4N]$ . Every coordinate of  $\varepsilon d$  likewise has a positive integer numerator at most  $4N$  when written with denominator  $4N$ . Consequently the reduced numerators and denominators of both outputs are at most  $4N$ .

*Proof.* Let  $B$  have the generators  $u_i, u_g$  as columns, so  $M = B^\top B$ . Set

$$c^* = (\mathbf{1}_n, \frac{1}{2}\mathbf{1}_p, \xi), \quad \xi = \frac{\eta^2 - p/4}{2\eta} = -\frac{1}{8\eta}.$$

Their inner products give

$$u_i^\top c^* = 2 + \frac{|S_i|}{2} = \frac{M_{ii}}{2}, \quad u_g^\top c^* = \frac{p}{4} + \eta\xi = \frac{M_{gg}}{2}.$$

Thus  $\delta = 2B^\top c^*$ . The matrix  $\Pi = B(B^\top B)^{-1}B^\top$  is the orthogonal projection onto  $\text{range}(B)$ , whence

$$s = 4 \|\Pi c^*\|^2 \leq 4 \|c^*\|^2 = 4n + p + 4\xi^2 = 4n + p + \frac{1}{4(p-1)} < \frac{N}{2}.$$

For the lower bound, Cauchy–Schwarz in the  $M$  inner product gives

$$M_{ii}^2 = (e_i^\top M M^{-1} \delta)^2 \leq (e_i^\top M e_i)(\delta^\top M^{-1} \delta) = M_{ii} s.$$

Since  $M_{ii} > 0$ , division proves the lower bound.

The binary diagonal equalities follow from  $\delta = \text{diag}(M)$ . The lower block  $\varepsilon M$  is positive definite, and the Schur complement of that block in  $Y_\varepsilon - \theta e_0 e_0^\top$  is  $1 - \theta - \varepsilon s > 0$  whenever  $\theta \leq 1/2$ . This proves (28).

The entries of  $4NY_\varepsilon$ , apart from its root entry  $4N$ , are

$$16 + 4|S_i|, \quad 2p - 1, \quad 4|S_i \cap S_j| \ (i \neq j), \quad 2|S_i|.$$

They are nonnegative integers. If a set index exists,  $n \geq 1$  and  $16 + 4|S_i| \leq 16 + 4p \leq 4N$ ; all other bounds follow from  $|S_i| \leq p$ . If  $n = 0$ , only the root and  $g$  entries remain, and the bound still holds. From (23), the numerators of  $\varepsilon d$  with denominator  $4N$  are

$$16 + 4|S_i|, \quad 2p - 1, \quad 32 + 4|S_i \triangle S_j| \ (i \neq j), \quad 2p + 15.$$

These are positive. Whenever a set index exists,  $4p + 32 \leq 4N$ , so they are at most  $4N$ ; without a set index only  $2p - 1$  remains. Reduction of a fraction cannot increase its numerator or denominator.  $\square$

The point  $c^*$  is also the sphere center in (8). Projecting  $c^*$  onto  $\text{range}(B)$  leaves the difference of squared distances in that identity unchanged. Thus the same sphere that explains the integer slack supplies the bound on  $s$  needed for a scaling with polynomially bounded numerator and denominator.

**Theorem 3.3** (Hard points in standard relaxations). *Suppose the instance is Exact Cover by 3-Sets (X3C):  $|S_i| = 3$ ,  $p = 3q$ , and  $q \geq 3$ . Let  $\hat{d} = \varepsilon d$  and  $Y_\varepsilon$  be as above.*

1. *The violated hypermetric inequalities of  $\hat{d}$  are exactly those of*

$$b_T = (2 - q)e_0 + \sum_{i \in T} e_i - e_g, \quad T \text{ an exact cover.}$$

*Each has gonality  $2q - 1$ , support size  $q + 2$ , and violation  $1/(2N)$ . All hypermetric inequalities of gonality at most  $2q - 3$  hold.*

2. *The violated rounded positive semidefinite inequalities are exactly those described by  $b_T$  and  $-b_T$ ; each pair describes the same inequality and has violation  $1/(2N)$ . No rounded positive semidefinite inequality with  $|\sigma(b)| \geq 3$  is violated.*

3. *The violated binary split inequalities at  $Y_\varepsilon$  are exactly*

$$v = (0, -x, 1) \quad \text{and} \quad v = (-1, x, -1), \quad x \in \{0, 1\}^n, \quad Ax = \mathbf{1}_p.$$

*They have  $q_{Y_\varepsilon}(v) = -1/(2N)$ .*

4. *The point  $\hat{d}$  belongs to  $\text{MET}(V)$ , and  $Z(\hat{d})$  belongs to the interior of  $\text{ELL}(V)$  relative to its affine hull. Its Gram matrix  $M(\hat{d}) = \varepsilon M$  and binary moment matrix  $Y_\varepsilon$  are positive definite, and  $0 < \hat{d}_{uv} < 1$  for distinct  $u, v$ .*

*All output numerators and denominators are at most  $4N$ .*

*Proof.* Summing the equations  $Ax = \mathbf{1}_p$  shows that every exact cover uses exactly  $q$  sets. By (24), all hypermetric violators are the stated  $b_T$ , with gonality  $|2 - q| + q + 1 = 2q - 1$ , support size  $q + 2$ , and  $Q(b_T, \hat{d}) = \varepsilon/2 = 1/(2N)$ .

To classify rounded positive semidefinite inequalities, take any integer vector  $b = (b_0, z)$  and write  $\sigma = \sigma(b)$ . By (6), for odd  $\sigma$  and  $u = (\sigma, -2z)$  the rounded positive semidefinite slack at  $\hat{d}$  is

$$\frac{\sigma^2 - 1}{4} - Q(b, \hat{d}) = \frac{\sigma^2 - 1}{4} + \varepsilon(z^\top M z - \sigma \delta^\top z) = \frac{u^\top Y_\varepsilon u - 1}{4}. \quad (29)$$

If  $|\sigma| \geq 3$ , (28) with  $\theta = 1/2$  yields  $u^\top Y_\varepsilon u > \sigma^2/2 \geq 9/2 > 1$ . If  $\sigma = 1$ , the slack is  $\varepsilon g_M(z)$ ; if  $\sigma = -1$ , it is  $\varepsilon g_M(-z)$ . Applying Theorem 3.1 gives exactly  $b_T$  and  $-b_T$ , with the stated violations.

For a split vector  $v$ , (16) expresses its slack in terms of  $2v + e_0$ . This vector has an odd root coordinate and even remaining coordinates, so (29) applies. The two possible violated parity vectors are  $(1, -2x, 2)$  and  $(-1, 2x, -2)$  for exact covers  $x$ , giving precisely the two stated split vectors and their slacks.

It remains to verify the relaxation promises. Positive definiteness and the encoding bound follow from Lemma 3.2. By the congruence (11),  $Z(\hat{d})$  is positive definite with unit diagonal. It lies in the interior of the ellipsope. Its  $2 \times 2$  principal minors are positive, so  $|1 - 2\hat{d}_{uv}| < 1$  for distinct  $u, v$ , proving the strict coordinate bounds.

Every triangle inequality is a hypermetric inequality of gonality three. Since  $q \geq 3$ , none is violated. Every perimeter inequality has vector  $e_u + e_v + e_w$  and sum three, so it holds by the rounded positive semidefinite classification. These are the defining inequalities of  $\text{MET}(V)$ .  $\square$

*Remark 3.4.* The rounded positive semidefinite and split classifications, the positive definiteness margin, and the elliptope conclusion hold for every exact-cover instance with  $p \geq 2$ . In a general instance, all triangle inequalities hold if and only if no exact cover uses at most two sets. Indeed, a one-set or two-set cover in (24) produces a triangle vector, while every cover with at least three sets produces a vector of gonality at least five. For X3C with  $q = 2$ , the hypermetric violators have gonality three and support size three; these points still satisfy the elliptope and coordinate promises.

**Corollary 3.5.** *Hypermetric violation, rounded positive semidefinite violation, and Boros–Hammer violation are strongly NP-complete. For every fixed  $K \geq 3$ , this remains true at rational distance vectors  $d \in \text{MET}(V)$  with  $Z(d) \succ 0$  that satisfy every hypermetric inequality of gonality at most  $2K - 3$ . The associated Gram and binary moment matrices may both be required to be positive definite. The reduction has output numerators and denominators bounded by  $4(8n + 2p + 1)$ , where  $n$  is the number of sets and  $p$  the number of elements of the padded X3C instance. Hypermetric cone membership is strongly coNP-complete.*

*Proof.* X3C is NP-complete [12]. For a fixed  $K \geq 3$ , add  $K$  disjoint triples of new elements and one set for each new triple. Each added set is forced, so exact covers of the original instance correspond bijectively to exact covers after padding. The padded instance has  $q \geq K$ , and its size grows by a constant depending only on  $K$ . By Theorem 3.3, it has an exact cover if and only if any one of the three inequality families has a violator at the constructed point. All the stated relaxation promises hold on both yes- and no-instances.

The source instance has an explicit finite universe. Computing the intersections in (19) takes polynomial time, and the output has  $O((n+2)^2)$  entries with numerators and denominators  $O(n+p)$ . Thus its encoding is polynomial even if the numerical data are encoded in unary, proving strong NP-hardness.

The short-certificate results Lemmas 5.3 and 5.4 prove NP membership for arbitrary rational inputs. For any rational distance vector, set  $M = M(d)$ ,  $\delta = \text{diag}(M)$ , and  $Y = Y(\delta, M)$ . The algebra in (29) applies with  $\varepsilon = 1$  and identifies a rounded positive semidefinite vector of odd sum with a split vector:  $u = (\sigma, -2z) = 2v + e_0$  and  $v = ((\sigma - 1)/2, -z)$ . This transformation preserves polynomial certificate length in both directions. Boros–Hammer inequalities are the binary moment form of those split inequalities. Taking the complement of hypermetric violation proves the cone-membership statement.

For hypermetric violation alone, homogeneity also permits the integral output  $4d$  in (23). Its coordinates are at most  $4p + 32$ , and its Gram matrix is positive definite. It satisfies all triangle inequalities and the same gonality promise, so this also gives strong hardness with integral distances.  $\square$

## 4 Consequences for named inequality families

The base construction transfers to several infinite families because its rounded psd violations are exactly its hypermetric violations. A second, orthogonal extension permits coefficients of absolute value at most one without sacrificing positive definiteness.

**Theorem 4.1.** *Separation of each of the following families is strongly NP-complete: hypermetric correlation inequalities, gap-1 inequalities, rounded psd inequalities, Boros–Hammer inequalities, and the union of the  $k$ -gonal families over all  $k$ . For every fixed  $K \geq 3$ , hardness holds at rational distance vectors  $d \in \text{MET}(V)$  with  $Z(d) \in \text{relint ELL}(V)$  that satisfy every rounded psd inequality of gonality at most  $2K - 3$ . Their Boolean quadric moment matrices are positive definite, and their coordinates have polynomially bounded numerators and a polynomially bounded common denominator.*

*Proof.* Use the points  $\varepsilon d$  of [Theorem 3.3](#), with  $q \geq K$ . Every hypermetric inequality is a gap-1 inequality: its coefficient sum is one, so its gap is at most one, while parity excludes zero. Every gap-1 inequality is rounded psd. At these points, the only rounded psd violations have coefficient sum  $\pm 1$  and correspond to exact covers. The members of the  $k$ -gonal union with even coefficient sum are psd inequalities and therefore hold. Thus each family has a violation if and only if an exact cover exists. The covariance map identifies hypermetric inequalities with hypermetric correlation inequalities and rounded psd inequalities with Boros–Hammer inequalities. The size bounds and relaxation properties follow from [Theorem 3.3](#).

For membership in NP, hypermetric correlation inequalities use [Lemma 5.3](#). Rounded-psd and Boros–Hammer inequalities use [Lemma 5.4](#) and their exact covariance and split correspondences. For the  $k$ -gonal union, an even-sum violation requires an indefinite  $Z = J - 2D$ . By [Lemma 5.1](#), it has an integral negative direction of polynomial encoding length; doubling that direction gives an even-sum certificate. An odd-sum violation is rounded psd.

For gap-1, suppose  $b$  is violated and choose  $a \in \{\pm 1\}^V$  with  $|a^\top b| = 1$ . After a possible global negation, the vector  $b' = \text{diag}(a)b$  has coefficient sum one. Switch  $d$  on  $S = \{u : a_u = -1\}$ , so that  $Z(d^S) = \text{diag}(a)Z(d)\text{diag}(a)$ . Then  $(b')^\top Z(d^S)b' < 1$ , equivalently  $Q(b', d^S) > 0$ . Conversely, a hypermetric violation on a switched input gives a gap-1 violation on the original input: the switching signs witness a signed sum of absolute value one, and odd parity proves that the gap is exactly one. A certificate consists of  $S$  and a short hypermetric witness for  $d^S$ . Its length is polynomial by [Lemma 5.3](#). This argument applies to arbitrary rational inputs, including inputs outside  $[0, 1]^{E(V)}$ .  $\square$

#### 4.1 Signed Boolean quadric families and their facets

Recall the signed family (17), whose slack is  $h_Y(\mathbf{1}_S - \mathbf{1}_T, s)/2$ . Its cut subfamily has  $s = 0$ ; its clique subfamily has  $T = \emptyset$  and  $0 \leq s \leq |S| - 1$ . These are the families (10)–(12) in Letchford [18]; the unrestricted Boros–Hammer family is (13). See also Padberg [21] and Letchford and Sørensen [19].

Complement one variable  $g$ : set

$$x'_g = 1 - x_g, \quad y'_{ig} = x_i - y_{ig} \quad (i \neq g),$$

and leave the other coordinates unchanged. Denote this map by  $\psi_g$ . Its moment matrix is  $AYA^\top$ , where  $A$  is the identity except that row  $g$  is  $e_0^\top - e_g^\top$ . Since  $A^2 = I$ , it preserves positive definiteness. If  $v'$  is  $v$  with its  $g$ -entry negated, then

$$h_{AYA^\top}(v, s) = h_Y(v', s - v_g). \quad (30)$$

Indeed, with  $\hat{u} = (-s, v)$ ,  $h_Y(v, s) = \hat{u}^\top Y \hat{u} - e_0^\top Y \hat{u}$ ,  $A^\top \hat{u} = (-(s - v_g), v')$ , and  $e_0^\top A = e_0^\top$ . The shift of  $s$ , as well as the sign change of  $v_g$ , is part of the switching rule. On the Boolean cube this map complements  $x_g$ , so it is an affine automorphism of the Boolean quadric polytope. On its cut image it is switching on  $\{g\}$ .

We prove the facet property needed below directly.

**Lemma 4.2.** *Let  $S \subseteq W$ ,  $t = |S| \geq 3$ , and  $1 \leq r \leq t - 2$ . The clique specialization of (17) with parameter  $r$  defines a facet of  $\text{BQP}(W)$ , including when  $S \neq W$ .*

*Proof.* At a binary point its slack is

$$F(x) = \sum_{\{i,j\} \subseteq S} x_i x_j - r \sum_{i \in S} x_i + \frac{r(r+1)}{2} = \frac{(\mathbf{1}_S^\top x - r)(\mathbf{1}_S^\top x - r - 1)}{2}.$$

Its tight points are exactly those with  $\mathbf{1}_S^\top x \in \{r, r+1\}$ . An affine equation on Boolean quadric coordinates is a multilinear polynomial of degree at most two on the Boolean cube. First suppose such a polynomial  $p = c + \sum_{i \in S} a_i x_i + \sum_{\{i,j\} \subseteq S} a_{ij} x_i x_j$  vanishes on both tight layers. For distinct  $i, j \in S$ , comparing the points supported on  $R \cup \{i\}$  and  $R \cup \{j\}$  gives

$$a_i - a_j + \sum_{k \in R} (a_{ik} - a_{jk}) = 0$$

for every  $R \subseteq S \setminus \{i, j\}$  of size  $r-1$  or  $r$ . For any  $k \notin \{i, j\}$ , choose  $R$  of size  $r-1$  excluding  $k$ ; this is possible because  $r \leq t-2$ . Subtract the equations for  $R$  and  $R \cup \{k\}$ . Thus  $a_{ik} = a_{jk}$ . All pair coefficients have a common value  $a$ , and all linear coefficients have a common value  $b$ . The two layer equations give  $b = -ra$  and  $c = r(r+1)a/2$ . Therefore  $p = aF$ .

Now allow variables outside  $S$ . Set them all to zero. The remaining polynomial is a multiple of  $F$ . Setting one outside variable to one shows that its coefficient, an affine function on  $S$ , vanishes on both layers. Such an affine function is zero: swaps show that all its linear coefficients agree, and the two distinct layer values then force that coefficient and the constant to vanish. Setting two outside variables to one eliminates their pair coefficient. Hence every affine equation on the tight points is a multiple of  $F$ . The Boolean quadric polytope has full dimension: evaluation at  $0, e_i, e_i + e_j$  shows that  $1, x_i, x_i x_j$  are linearly independent on the Boolean cube. Its tight points therefore span a face of codimension one.  $\square$

**Theorem 4.3.** *Separation of Padberg's cut inequalities, Padberg's clique inequalities, and (17) is strongly NP-complete. For every fixed  $K \geq 3$ , hardness holds at positive definite Boolean quadric moment points whose cut images  $d$  satisfy  $d \in \text{MET}(V)$ ,  $Z(d) \in \text{relint ELL}(V)$ , and every rounded psd inequality of gonality at most  $2K-3$ . Their coordinates have a polynomially bounded common denominator and polynomially bounded numerators. Every violated inequality in these constructions defines a facet.*

*Proof.* At the point  $Y_\varepsilon$  of Theorem 3.3, (15) and the exact list of split violations give precisely the violated Boros–Hammer data

$$(v, s) = (\mathbf{1}_C - e_g, 0), \quad (-\mathbf{1}_C + e_g, -1),$$

for the exact covers  $C$ . These are two forms of the same inequality. Consequently, up to  $(v, s) \sim (-v, -s-1)$ , the violated members of (17) are exactly  $S = C, T = \{g\}, s = 0$ . They are cut inequalities, with violation  $1/(4N)$ .

At  $\psi_g(Y_\varepsilon)$ , (30) gives exactly

$$(v, s) = (\mathbf{1}_C + e_g, 1), \quad (-\mathbf{1}_C - e_g, -2).$$

Thus the violated clique inequalities are exactly  $S = C \cup \{g\}, s = 1$ , again with violation  $1/(4N)$ . They are also the only violated members of (17), up to the same equivalence. Their supports have size  $q+1$ , so Lemma 4.2 applies. The cut violators are their images under  $\psi_g$  and are therefore facets too.

The switched moment matrix is positive definite by congruence. Its changed entries are

$$x'_g = 1 - \frac{2p-1}{4N}, \quad y'_{ig} = \frac{11}{2N}$$

for X3C, so its coordinates are in  $[0, 1]$  with common denominator  $4N$ . Cut switching is diagonal-sign congruence on  $Z$  and preserves the gonality of rounded psd inequalities. It permutes triangle and

perimeter inequalities, hence preserves the metric polytope as well as the ellipsope interior. All promised properties follow from [Theorem 3.3](#). For a fixed  $K$ , the X3C restriction  $q \geq K$  remains strongly NP-hard, so these encodings establish the claimed strong hardness.

For membership in NP, a cut inequality is specified by  $(S, T)$ . A clique inequality is specified by  $S$  and an integer in a range of length  $|S|$ . For [\(17\)](#), fix  $(S, T)$  and put  $\ell = x(S) - x(T)$ . Its violation as a function of  $s$  is  $-s^2/2 + s(\ell - 1/2) + c$ , for a constant  $c$ . Its maximum over the integers is attained at  $s = \lfloor \ell \rfloor$ , which has polynomial bit length. Thus  $(S, T)$  alone suffices as a certificate: the verifier computes this parameter and the slack.  $\square$

## 4.2 Pure hypermetric and cut-form clique families

The odd-clique family in [Table 1](#) is precisely the set of switchings of the all-positive clique inequalities

$$\sum_{\{u,v\} \subseteq S} d_{uv} \leq \frac{|S|^2 - 1}{4}, \quad |S| \text{ odd}, \quad (31)$$

usually called the Barahona–Mahjoub clique inequalities; see Barahona and Mahjoub [\[3\]](#), Galli et al. [\[11\]](#), Letchford [\[18\]](#).

Take the X3C construction with  $q \geq 3$ , and put

$$m = q - 2, \quad \tau^2 = \frac{1}{8q}, \quad \widetilde{M} = M \oplus \tau^2 I_m.$$

Add indices  $o_1, \dots, o_m$  and let  $\widetilde{V} = V \cup \{o_1, \dots, o_m\}$ . Its covariance distance  $\widetilde{d} = d(\widetilde{M})$  satisfies

$$\widetilde{d}_{0o_j} = \tau^2, \quad \widetilde{d}_{o_j o_k} = 2\tau^2 \quad (j \neq k), \quad \widetilde{d}_{o_j u} = d_{0u} + \tau^2 \quad (u \in W).$$

In a Gram representation, the new points are the root moved by  $\tau$  in distinct new orthogonal directions. They are distinct from each other and from the original points. Let  $\widetilde{Y}_\varepsilon$  be the scaled moment matrix of this extension, with the original  $N = 8n + 2p + 1$  and  $\varepsilon = 1/N$ . Let  $d''$  be  $\varepsilon \widetilde{d}$  switched on  $U = \{g, o_1, \dots, o_m\}$ .

**Theorem 4.4.** *Separation of pure hypermetric inequalities, odd-clique inequalities, and the all-positive clique inequalities (31) is strongly NP-complete. For every fixed  $K \geq 3$ , hardness holds at rational distance vectors  $d \in \text{MET}(V)$  with  $Z(d) \in \text{relint ELL}(V)$  that satisfy every rounded psd inequality of gonality at most  $2K - 3$ . Their moment and covariance Gram matrices are positive definite, and their coordinates have polynomially bounded numerators and common denominator.*

*More precisely, at  $\varepsilon \widetilde{d}$  the pure hypermetric violators for a cover  $C$  are exactly*

$$b^* = (0, \mathbf{1}_C, -1, -\mathbf{1}_m), \quad b^{(j)} = (-1, \mathbf{1}_C, -1, -\mathbf{1}_m + e_j), \quad 1 \leq j \leq m, \quad (32)$$

*in the order  $(0, 1, \dots, n, g, o_1, \dots, o_m)$ . The odd-clique violators are these vectors and their negatives. At  $d''$ , the violated all-positive clique inequalities are exactly those with  $S = C \cup U$  for the covers  $C$ . All these inequalities define facets. If a cover exists, the maximum positive violation is  $(q + 3)/(4qN)$  for pure hypermetric and odd-clique inequalities at  $\varepsilon \widetilde{d}$ , and  $(q + 2)/(4qN)$  for [\(31\)](#) at  $d''$ . These maxima are zero if there is no cover.*

*Proof.* The block matrix  $\widetilde{M}$  is positive definite. For integral  $z \in \mathbb{Z}^W$  and  $w \in \mathbb{Z}^m$ ,

$$g_{\widetilde{M}}(z, w) = g_M(z) + \tau^2 \sum_{j=1}^m w_j(w_j - 1). \quad (33)$$

The added term is nonnegative. By [Theorem 3.1](#), a hypermetric violation is therefore possible exactly when  $z = (\mathbf{1}_C, -1)$  for a cover  $C$  and  $\sum_j w_j(w_j - 1) < 4q$ . Its coefficient vector is

$$b = (2 - q - \mathbf{1}^\top w, \mathbf{1}_C, -1, w). \quad (34)$$

Its gonality is at least  $2q - 1$ , because

$$|2 - q - \mathbf{1}^\top w| + \sum_j |w_j| \geq q - 2.$$

Write  $s_M = \delta^\top M^{-1} \delta$ . The extended Schur complement parameter is  $s_{\tilde{M}} = s_M + m\tau^2$ . By [Lemma 3.2](#),

$$\varepsilon s_{\tilde{M}} < \frac{1}{2} + \frac{1}{8N} < \frac{3}{4}.$$

Consequently both  $\tilde{Y}_\varepsilon$  and  $\tilde{Y}_\varepsilon - e_0 e_0^\top / 4$  are positive definite. By (11),  $Z(\varepsilon \tilde{d}) = L \tilde{Y}_\varepsilon L^\top$ . If  $b = (b_0, z)$ , then  $L^\top b = (\sigma(b), -2z)$ . Hence  $b^\top Z b > \sigma(b)^2 / 4$  for nonzero  $b$ . This excludes rounded psd violations with  $|\sigma(b)| \geq 3$ . The remaining rounded psd violators have coefficient sum  $\pm 1$  and are the vectors (34) and their negatives. Thus none has gonality at most  $2q - 3$ . For  $q \geq 3$ , triangle inequalities have gonality three and perimeter inequalities have coefficient sum three, so the extended point is metric feasible. Its  $Z$  matrix is positive definite, and its unit diagonal implies that all distances lie strictly between zero and one.

For a pure hypermetric violator, let  $a$  entries of  $w$  be  $+1$  and  $k$  be  $-1$ . Its root coefficient has absolute value at most one, so  $k - a \geq q - 3 = m - 1$ , while  $k + a \leq m$ . Therefore  $a = 0$  and  $k \in \{m - 1, m\}$ . This gives precisely (32). All these vectors meet the bound  $\sum_j w_j(w_j - 1) < 4q$ . The violations are

$$Q(b^*, \varepsilon \tilde{d}) = \frac{q + 2}{4qN}, \quad Q(b^{(j)}, \varepsilon \tilde{d}) = \frac{q + 3}{4qN}.$$

Every odd-clique violator must have coefficient sum  $\pm 1$ , by the preceding exclusion, so its global sign can be chosen to make it pure. This proves the exact lists and their maximum.

Let  $\Sigma$  be diagonal with entries  $-1$  on  $U$  and  $+1$  elsewhere. Then  $Z(d'') = \Sigma Z(\varepsilon \tilde{d}) \Sigma$ . The map  $b \mapsto \Sigma b$  preserves odd parity, gonality, the  $\{0, \pm 1\}$  restriction, and rounded psd violation. For the vector (34), its switched vector is

$$\Sigma b = (2 - q - \mathbf{1}^\top w, \mathbf{1}_C, 1, -w).$$

It has all nonzero entries positive exactly when  $w \in \{0, -1\}^m$  and  $2 - q - \mathbf{1}^\top w \in \{0, 1\}$ . If  $k$  entries of  $w$  are  $-1$ , these conditions give  $2 - q + k \in \{0, 1\}$  and  $k \leq q - 2$ , so  $k = q - 2$ . Thus  $w = -\mathbf{1}_m$ , the root entry is zero, and the support is  $C \cup U$ . The globally negated vectors cannot give an all-positive support: their coordinates on the nonempty cover  $C$  remain negative after switching on  $U$ . Conversely, this choice is violated. Its support size is  $2q - 1$  and its violation is  $(q + 2)/(4qN)$ , establishing the exact all-positive list.

Every coefficient vector just listed has odd support size  $2q - 1 \geq 5$ . An all-positive clique inequality on an odd support of size  $t$  not containing the root is the covariance image of the clique specialization of (17) with  $s = (t - 1)/2$ . If its support contains the root, it is the covariance image with  $t - 1$  variables and  $s = (t - 1)/2$ . Both meet the range in [Lemma 4.2](#) when  $t \geq 5$ . The covariance map is a linear bijection of the Boolean quadric and cut polytopes, as is seen directly on their binary vertices and convex hulls. It preserves facets, as do cut switchings. This proves all claimed facet properties, including those of the maximizing vectors  $b^{(j)}$ .

The entries of  $8qN \tilde{Y}_\varepsilon$  are integral and lie between zero and  $8qN$ . The distances and their switchings likewise have denominator dividing  $8qN$  and lie in  $(0, 1)$ , so their numerators are bounded



by  $8qN$ . Switching preserves the metric polytope, elliptope interior, and absence of rounded psd violations of gonality at most  $2q - 3$ ; its moment matrix is positive definite by the inverse congruence with  $L$ . Its principal covariance Gram matrix is therefore positive definite as well. The construction has polynomial size. Taking  $q \geq K$  proves strong hardness with each stated promise. Membership in NP is immediate: the certificates have entries in  $\{0, \pm 1\}$  or specify a subset.  $\square$

*Remark 4.5.* If the all-positive clique family includes even supports with right-hand side  $\lfloor |S|^2/4 \rfloor$ , those members are psd inequalities and hold at  $d''$ . The hardness result therefore holds under either convention. At  $\varepsilon \tilde{d}$ , the maximum positive hypermetric and rounded psd violation, conditional on a cover existing, is  $1/(2N)$ : in (33), the added term vanishes exactly when  $w \in \{0, 1\}^m$ . These unnormalized maxima describe the construction; they do not give a lower bound on the violation needed for hardness.

## 5 Certificates and algorithmic boundaries

All algorithms in this section take exact rational input. The input length includes the binary encodings of its numerators and denominators. A positive semidefinite matrix of small rank need not have a rational Gram factor with the same row dimension as its rank, so we work with a rational quotient and its rational inner product.

### 5.1 Rational linear algebra and short certificates

**Lemma 5.1.** *For a rational symmetric matrix  $A$ , polynomial-time exact linear algebra either certifies  $A \succeq 0$  and finds a positive definite principal submatrix  $A_{KK}$  of order  $\text{rank } A$ , or returns an integer vector  $y$  of polynomial encoding length with  $y^\top A y < 0$ .*

*Proof.* Apply symmetric elimination to the current Schur complement  $C$ . A negative diagonal entry gives a negative direction  $e_i$ . If  $C_{ii} = 0$  and  $C_{ij} \neq 0$ , take  $z_j = 1$  and  $z_i = -(C_{jj} + 1)/(2C_{ij})$ , with all other entries zero; then  $z^\top C z = -1$ . Otherwise pivot on a positive diagonal entry. For a partition with that entry first, a direction  $z_R$  in the new Schur complement lifts by setting  $z_i = -C_{iR}z_R/C_{ii}$ , without changing its quadratic value. If the remaining matrix is zero, the matrix is positive semidefinite, and the positive pivots identify the required set  $K$ .

After pivots indexed by  $K$ , the Schur-complement entries are ratios of minors of the original matrix. Clearing input denominators and applying the determinant bound shows that these ratios have polynomial encoding length. A negative direction can be lifted at once by  $z_K = -A_{KK}^{-1}A_{KR}z_R$ , where  $z_R$  has at most two nonzero entries. Cramer's rule gives polynomial encoding length for the lifted direction. Multiplying by its common denominator gives the asserted integer vector. Fraction-free elimination, or rational elimination with reduced fractions, performs these operations in polynomial time.  $\square$

**Lemma 5.2.** *Let  $A \succeq 0$  be rational of rank  $r > 0$ . In polynomial time one can find  $K$ ,  $P = A_K$ , and  $G = A_{KK}^{-1} \succ 0$  such that*

$$A = P^\top G P,$$

*and a rational basis  $R_\Lambda$  of the rank- $r$  lattice  $\Lambda = P\mathbb{Z}^m$ , together with an integer matrix  $T$  of polynomial encoding length satisfying  $R_\Lambda = PT$ . If a rational vector  $y$  of polynomial encoding length belongs to  $\Lambda$ , an integer solution of  $Pz = y$  of polynomial encoding length can be found in polynomial time.*



*Proof.* Choose  $K$  by Lemma 5.1. The Schur complement of  $A_{KK}$  is zero because  $A$  has rank  $r$ , which gives  $A = P^\top GP$ . Let  $D$  be a common denominator of  $P$ . Its encoding length is at most the sum of the encoding lengths of the denominators of  $P$ . The integer matrix  $DP$  has full row rank. Its column Hermite normal form gives a basis  $H$  of  $(DP)\mathbb{Z}^m$  and solves integral systems with this coefficient matrix in polynomial time, with polynomial encoding lengths [14]. Take  $R_\Lambda = H/D$ , and solve  $(DP)t_j = H_{\cdot j}$  for each basis column to obtain  $T$ .

Thus  $\Lambda = R_\Lambda \mathbb{Z}^r$  is a lattice in  $\mathbb{R}^r$ . For any  $y \in \Lambda$ , solving  $(DP)z = Dy$  gives the final assertion. Here  $Dy$  is integral; all input and output lengths in this system are polynomial. This argument also explains why linear dependence among the columns of a real Gram factor causes no difficulty.  $\square$

**Lemma 5.3.** *For any rational symmetric  $M$ , if  $g_M(z) < 0$  for some  $z \in \mathbb{Z}^m$ , there is such a vector of encoding length polynomial in the encoding length of  $M$ . Consequently, hypermetric violation has polynomial certificates.*

*Proof.* Put  $\delta = \text{diag } M$ . If  $M$  is not positive semidefinite, take the integer negative direction from Lemma 5.1, and choose its sign so that  $\delta^\top z \geq 0$ . Then  $g_M(z) = z^\top Mz - \delta^\top z < 0$ .

Suppose next that  $M \succeq 0$  but  $\delta \notin \text{range } M$ . Since  $\text{range } M = (\ker M)^\perp$ , a rational kernel basis contains a vector  $k$  with  $\delta^\top k \neq 0$ . Exact elimination supplies such a vector with polynomial encoding length. Clear denominators and choose its sign so that  $\delta^\top k > 0$ . Then  $g_M(k) < 0$ .

It remains to consider  $M \succeq 0$  and  $\delta \in \text{range } M$ . If  $M = 0$ , then  $\delta = 0$  and there is no violator. Otherwise use Lemma 5.2, writing  $D_0 = M_{KK}$ ,  $P = M_K$ , and  $G = D_0^{-1}$ . The range condition gives  $\delta = P^\top G \delta_K$ . With  $y = Pz$  and  $t = \delta_K/2$ ,

$$g_M(z) = (y - t)^\top G(y - t) - t^\top Gt. \quad (35)$$

A violator therefore satisfies  $\|y\|_G < 2\|t\|_G$ , where  $\|a\|_G^2 = a^\top Ga$ . By Cauchy–Schwarz,

$$|y_i| \leq \sqrt{(D_0)_{ii}} \|y\|_G < 2\sqrt{(D_0)_{ii}} \|t\|_G.$$

The square of the bound is rational of polynomial encoding length, so its magnitude is at most  $2^{\text{poly}(L)}$ , where  $L$  is the input length. Moreover, every coordinate of  $y = Pz$  has denominator dividing a common denominator of  $P$ . Hence  $y$  itself has polynomial encoding length. Lemma 5.2 gives a polynomial-size integer  $z'$  with  $Pz' = y$ . Equation (35) gives  $g_M(z') = g_M(z) < 0$ . The corresponding hypermetric vector is  $(1 - \mathbf{1}^\top z', z')$ , whose encoding length is also polynomial.  $\square$

For general integer moments, let  $Y = \begin{bmatrix} 1 & x^\top \\ x & B \end{bmatrix}$  be rational and symmetric. No Boolean diagonal identities are assumed unless stated. We use the parity identity (16), with  $u = e_0 + 2v$ .

**Lemma 5.4.** *If a rational symmetric  $Y$  with  $Y_{00} = 1$  violates an integer split, then it has a violating integer split vector of polynomial encoding length. Thus unrestricted integer split violation belongs to NP.*

*Proof.* If  $Y$  is not positive semidefinite, take an integer negative direction  $v$  from Lemma 5.1 and choose its sign so that  $v^\top Y e_0 \leq 0$ . Then  $q_Y(v) < 0$ .

Otherwise apply Lemma 5.2 to  $Y$ . A violated split gives  $u = e_0 + 2v$  with  $u^\top Y u < 1$ . Set  $y = Pu$ . Since  $Y = P^\top GP$ ,  $y^\top G y < 1$ , and Cauchy–Schwarz gives

$$|y_i| < \sqrt{(Y_{KK})_{ii}}.$$

The common denominator of  $P$  bounds the denominators of  $y$ , so  $y$  has polynomial encoding length. The integral system  $2Pv' = y - Pe_0$  is feasible because  $v$  solves it. After clearing denominators, Hermite normal form produces a polynomial-size integer solution  $v'$ . Then  $P(e_0 + 2v') = y$ , and (16) gives  $q_Y(v') < 0$ .  $\square$

The positive definite binary points in [Corollary 3.5](#) are also general integer moment inputs. Their reduction, together with [Lemma 5.4](#), proves strong NP-completeness of unrestricted integer split violation.

## 5.2 Fixed rank, support and gonality

**Theorem 5.5.** *Hypermetric separation for rational  $M$  of rank  $r$ , and integer split separation for rational  $Y$  of rank  $r$  with  $Y_{00} = 1$ , can be solved in time  $r^{O(r)} \text{poly}(L)$ , where  $L$  is the input encoding length. The algorithm either returns a violated inequality or certifies that none exists; rank-zero inputs are decided directly.*

*Proof.* For hypermetric separation the first two cases in the proof of [Lemma 5.3](#) are polynomial-time constructive. In the remaining case, (35) asks whether the distance from  $t$  to  $\Lambda = P\mathbb{Z}^m$ , in the inner product  $G$ , is less than  $\|t\|_G$ . Construct  $R_\Lambda = PT$  by [Lemma 5.2](#). The problem is exact closest vector in dimension  $r$ , with rational Gram matrix  $F = R_\Lambda^\top G R_\Lambda$  and rational target coordinates  $\xi = R_\Lambda^{-1}t$ . To give a rational Euclidean instance, compute a rational factorization  $F = L_0 \text{diag}(d_i) L_0^\top$  with  $d_i = p_i/q_i > 0$ . The integer  $p_i q_i$  is a sum of polynomially many integer squares: in its binary expansion an even-position bit is one square, and an odd-position bit is two equal squares. Dividing the square roots by  $q_i$  expresses  $d_i$  as a sum of rational squares. If  $d_i = \sum_j \alpha_{ij}^2$ , take the rows of  $C$  to be  $\alpha_{ij} e_i^\top L_0^\top$ . This rectangular rational matrix has polynomial encoding length and  $C^\top C = F$ . Thus our problem is Euclidean closest vector to  $C\xi$  in the rank- $r$  lattice  $C\mathbb{Z}^r$ . The exact fixed-dimensional closest-vector algorithm takes  $r^{O(r)} \text{poly}(L)$  time [[13](#), Theorem 4.5]. Its lattice steps use inner products, projections and squared norms of the  $r$  generators. These can equally be computed from their rational Gram matrix; a larger ambient dimension affects only the polynomial cost of forming inner products. A returned coefficient vector  $a \in \mathbb{Z}^r$  lifts to the hypermetric witness  $z = Ta$ .

For split separation, non-positive-semidefinite inputs are handled by [Lemma 5.1](#). Otherwise put  $y_0 = P e_0$ . Then  $y_0^\top G y_0 = Y_{00} = 1$ , and

$$q_Y(v) < 0 \iff \|Pv + y_0/2\|_G^2 < \frac{1}{4}.$$

This is exact closest vector to  $-y_0/2$  in the same rank- $r$  lattice. A returned coefficient vector  $a$  yields  $v = Ta$ . All quotient data have polynomial encoding length. A nearest lattice point has norm at most twice the target norm, so its rational coordinates, and then its integer basis coordinates, have polynomial encoding length as well. Exact comparison treats equality at the threshold as satisfaction, as required by strict separation.  $\square$

**Corollary 5.6.** *For every fixed  $k$ , hypermetric and rounded psd separation restricted to coefficient vectors supported on at most  $k$  vertices is polynomial. Integer split separation restricted to  $|\text{supp } w| \leq k$  is also polynomial, with no bound on the integer coefficients. For every fixed  $k$ , separation over hypermetric or rounded psd inequalities of gonality at most  $k$  is polynomial.*

*Proof.* For hypermetrics, enumerate all nonempty vertex sets of size at most  $k$  and apply [Theorem 5.5](#) to each induced distance vector, choosing any vertex of that set as root. Its correlation matrix has order at most  $k - 1$ , so the running time per set is polynomial. Every supported hypermetric inequality belongs to one such induced instance. This avoids any need for a bound on facet coefficients. For rounded psd inequalities, apply the split correspondence to the Boolean moment matrix of each induced distance vector and use the split part of [Theorem 5.5](#); the matrix has order at most  $k$ .

For splits, enumerate sets  $I$  of at most  $k$  variable indices and apply [Theorem 5.5](#) to  $Y_{\{0\} \cup I, \{0\} \cup I}$ . Its rank is at most  $k + 1$ . There are  $O(n^k)$  such sets.

Finally, a coefficient vector with  $\sum_i |b_i| \leq k$  is a sum of at most  $k$  signed unit vectors. Enumerating these descriptions gives  $O((2|V| + 1)^k)$  candidates; filter by the defining parity or sum condition and evaluate each inequality exactly. These support and gonality bounds give polynomial algorithms for fixed parameters. Their displayed dependence on the number of vertices is not a fixed-parameter tractability claim for support or gonality.  $\square$

### 5.3 A fixed normalized threshold

Put  $S = B - xx^\top$  and define the normalized violation of a nonzero direction  $w$  and split vector  $v = (v_0, w)$  as  $-q_Y(v) / \|w\|^2$ . The denominator is the squared Euclidean norm of the variable direction.

**Lemma 5.7.** *For  $t = w^\top x$ , write  $\{t\} = t - \lfloor t \rfloor$  for its fractional part and put  $\phi(t) = \{t\}(1 - \{t\})$ . Then*

$$\max_{v_0 \in \mathbb{Z}} (-q_Y(v_0, w)) = \phi(t) - w^\top S w.$$

*If  $Y \succeq 0$ , this quantity is at most  $1/4$ .*

*Proof.* Expansion gives  $q_Y(v_0, w) = (v_0 + t)(v_0 + t + 1) + w^\top S w$ . Writing  $f = \{t\}$ , the minimum of the first term is  $-f(1 - f)$ , attained at  $v_0 = -\lfloor t \rfloor - 1$ . When  $f = 0$ , the offset  $v_0 = -\lfloor t \rfloor$  also attains it. If  $Y \succeq 0$ , its Schur complement  $S$  is positive semidefinite.  $\square$

**Theorem 5.8.** *Let  $Y \succeq 0$  be rational with  $Y_{00} = 1$ , and let  $\rho > 0$  be rational. An exact algorithm finds a split with normalized violation greater than  $\rho$ , or certifies that none exists, in time*

$$f(B_\rho) n^{B_\rho} \text{poly}(L + \text{size}(\rho)), \quad B_\rho = \max \left\{ 0, \left\lceil \frac{1}{4\rho} \right\rceil - 1 \right\}.$$

*In particular the problem is polynomial for each fixed positive  $\rho$ . The bound is  $XP$  in the inverse threshold.*

*Proof.* By [Lemma 5.7](#), any qualifying direction satisfies

$$0 < \|w\|^2 < \frac{1}{4\rho}, \quad \text{hence} \quad \|w\|^2 \leq B_\rho.$$

If  $B_\rho = 0$ , there is no qualifying direction. Otherwise enumerate all integer directions with this bound. Such a direction has at most  $B_\rho$  nonzero entries, each of magnitude at most  $\lfloor \sqrt{B_\rho} \rfloor$ . The number of candidates is at most

$$\sum_{j=1}^{\min(n, B_\rho)} \binom{n}{j} (2\lfloor \sqrt{B_\rho} \rfloor)^j \leq (B_\rho + 1)(2n\sqrt{B_\rho})^{B_\rho}.$$

For each candidate compute the best offset by [Lemma 5.7](#), and compare  $\phi(w^\top x) - w^\top S w$  with  $\rho \|w\|^2$  in exact rational arithmetic. The candidate coefficients have  $O(\log(B_\rho + 1))$  bits; the dot products, floor, fractional part and comparison have polynomial bit complexity in their input lengths. The best offset also has polynomial encoding length. Thus this finite search is exact, including at equality with the threshold.  $\square$

The same argument bounds any adaptive search after a positive normalized violation has been found. The theorem requires positive semidefinite input. A polynomial algorithm that finds an arbitrary split at a non-positive-semidefinite input does not by itself solve a normalized threshold problem there. Nor does the theorem establish an approximation hardness result for normalized violation.

**Proposition 5.9.** *For rational  $Y \succeq 0$  with  $Y_{00} = 1$ , the maximum raw split violation  $\mu(Y) = \max_{v \in \mathbb{Z}^{n+1}} (-q_Y(v))$  is attained and lies in  $[0, 1/4]$ . It equals  $1/4$  if and only if the integer kernel of  $Y$  meets  $e_0 + 2\mathbb{Z}^{n+1}$ . This equality condition is decidable in polynomial time.*

*Proof.* The parity identity gives  $q_Y(v) \geq -1/4$ , and  $v = 0$  gives zero. For a common denominator  $D$  of  $Y$ , the nonnegative values  $u^T Y u$ ,  $u \in e_0 + 2\mathbb{Z}^{n+1}$ , belong to  $D^{-1}\mathbb{Z}_{\geq 0}$ . This nonempty discrete set has a minimum, proving attainment. The maximum is  $1/4$  exactly when  $u^T Y u = 0$  for some such  $u$ . Positive semidefiniteness makes this equivalent to  $Y u = 0$ . After clearing denominators, Hermite normal form decides feasibility of the integral system  $2Y v = -Y e_0$  in polynomial time and supplies a split attaining  $1/4$  whenever the system is feasible.  $\square$

**Corollary 5.10.** *Unless  $P = NP$ , the maximum raw split violation at rational positive definite Boolean moment points has no polynomial-time multiplicative approximation with any finite factor. This means a value  $\hat{\mu}$  satisfying  $\mu(Y)/\alpha \leq \hat{\mu} \leq \alpha\mu(Y)$  for finite  $\alpha \geq 1$ , including when  $\mu(Y) = 0$ . There is also no polynomial-time algorithm guaranteed to approximate this value with absolute error smaller than  $1/(4N)$  on the points  $Y_\epsilon$  of [Theorem 3.3](#).*

*Proof.* The complete split classification in [Theorem 3.3](#) gives  $\mu(Y_\epsilon) = 1/(2N)$  when the X3C instance has a cover, and  $\mu(Y_\epsilon) = 0$  otherwise. A finite-factor guarantee distinguishes these cases by whether  $\hat{\mu}$  is positive. An absolute error strictly smaller than  $1/(4N)$  distinguishes them by comparison with  $1/(4N)$ . The input and this requested accuracy have polynomial encoding length. Both algorithms would decide X3C. The reduction does not give hardness at a fixed additive tolerance.  $\square$

An integer direction  $w \neq 0$  is primitive if the greatest common divisor of the absolute values of its coefficients is one.

**Lemma 5.11.** *Let  $w = kw'$  with  $0 \neq w' \in \mathbb{Z}^n$  and integer  $k \geq 2$ , and let  $v = (-s-1, w)$  with  $s \in \mathbb{Z}$ . Put*

$$\begin{aligned} t &= \left\lfloor \frac{2s+1-k}{2k} \right\rfloor, & b &= \frac{(2s+1)k - (2t+1)k^2}{2}, \\ a &= k^2 - b, & c &= (k(t+1) - s)(k(t+1) - s - 1). \end{aligned}$$

*Then  $a, b, c \geq 0$ ,  $a + b = k^2$ , and for every symmetric  $Y$ ,*

$$q_Y(-s-1, kw') = aq_Y(-t-1, w') + bq_Y(-t-2, w') + cY_{00}.$$

*If  $Y_{00} \geq 0$ , one of the two component splits has normalized violation at least that of  $v$ . In particular primitive directions suffice for violation detection and normalized optimization.*

*Proof.* Write  $z = (w')^T Y_{\{1, \dots, n\}, 0}$  and  $Z = (w')^T Y_{\{1, \dots, n\}, \{1, \dots, n\}} w'$ . The left-hand side is  $k^2 Z - (2s+1)kz + s(s+1)Y_{00}$ . Matching the coefficients of  $Z$  and  $z$  with the two component slacks gives the stated  $a$  and  $b$ . To obtain the remaining constant, apply the same identity to  $g(z) = (kz - s)(kz - s - 1)$  and  $f_j(z) = (z - j)(z - j - 1)$ . The quadratic and linear coefficients already match, and both  $f_t(t+1)$  and  $f_{t+1}(t+1)$  vanish. Thus the constant is  $c = g(t+1)$ . It is nonnegative because it is a

product of two consecutive integers. The definition of  $t$  gives  $(2t + 1)k \leq 2s + 1 < (2t + 3)k$ , hence  $0 \leq b < k^2$  and  $a > 0$ .

If  $Y_{00} \geq 0$ , negate the identity and divide by  $k^2 \|w'\|^2$ . The result is at most the convex combination, with weights  $a/k^2$  and  $b/k^2$ , of the component normalized violations. At least one component is therefore no smaller. Repeating with the greatest common divisor of the direction coefficients yields a primitive direction. The condition on  $Y_{00}$  is needed for this domination conclusion; the algebraic identity itself has no such restriction.  $\square$

## 5.4 Rank-one examples for general integer moments

The next results concern general integer moments. They do not describe fractional rank-one Boolean moments: if  $Y$  has rank one and also  $Y_{ii} = Y_{0i}$ , then  $x_i^2 = x_i$ , so  $x \in \{0, 1\}^n$  and every split is satisfied.

**Proposition 5.12.** *Every rational symmetric rank-one  $Y$  with  $Y_{00} = 1$  equals  $(1, x)(1, x)^\top$ . Let  $D$  be the least common denominator of the coordinates of  $x$ , and put  $p = D(1, x) \in \mathbb{Z}^{n+1}$ . Then  $\gcd(p) = 1$  and*

$$q_Y(v) = \frac{m(m + D)}{D^2}, \quad m = p^\top v.$$

*The point violates a split if and only if  $D \geq 2$ . Its maximum raw violation is  $\lfloor D^2/4 \rfloor / D^2$ , and a maximizing split can be found in polynomial time.*

*Proof.* Rank one and  $Y_{00} = 1$  imply  $Y = Y_0 Y_0^\top$ . For any prime dividing  $D$ , some reduced coordinate denominator contains its full power in  $D$ , so the corresponding coordinate of  $p$  is not divisible by that prime. Hence  $p$  is primitive, and  $p^\top v$  ranges over all integers. The displayed formula follows by expansion. It is negative exactly when  $-D < m < 0$ , and its minimum is attained at  $m = -\lfloor D/2 \rfloor$  or  $m = -\lceil D/2 \rceil$ . The extended Euclidean algorithm finds a polynomial-size integer vector  $a$  with  $p^\top a = 1$ ; multiplying  $a$  by the desired  $m$  gives a maximizing split. The least common denominator has polynomial encoding length.  $\square$

At fractional rational rank-one points this raw maximum is at least  $2/9$ , even if  $x$  is arbitrarily close to an integer vector. For example, with one variable  $x_1 = 1 - 1/D$ , the elementary direction has raw violation  $(D - 1)/D^2$ , whereas  $v = (-a, a e_1)$ ,  $a = \lfloor D/2 \rfloor$ , attains the raw maximum. For  $D \geq 4$  its normalized violation is  $(D - a)/(D^2 a) < (D - 1)/D^2$ , so it is strictly smaller than that of the elementary direction. This gives a reason to distinguish raw violation from normalized violation; it is not a statement about which choice gives the best objective bound.

**Proposition 5.13.** *At rational rank-one general integer moment points, deciding whether there exists a violated split vector  $v \in \{0, 1\}^{n+2}$  is NP-complete, although unrestricted split separation is polynomial there.*

*Proof.* Given a SUBSET SUM instance with positive integers  $a_1, \dots, a_n, T$  [15], set  $x_i = -12a_i/5$  and  $x_{n+1} = (12T - 2)/5$ , and take  $Y = (1, x)(1, x)^\top$ . For a 0/1 vector  $v$  put  $A_I = \sum_{i: v_i=1} a_i$ . The split is violated if and only if  $-1 < \tau < 0$ , where  $\tau = v_0 + \sum_i v_i x_i$ . If  $v_{n+1} = 0$ , then  $\tau = v_0 - 12A_I/5$ ; it is nonnegative when  $A_I = 0$  and at most  $-7/5$  otherwise. If  $v_{n+1} = 1$ , then

$$\tau = v_0 - \frac{2}{5} + \frac{12}{5}(T - A_I).$$

For  $T = A_I$ , the choice  $v_0 = 0$  gives  $\tau = -2/5$ . For  $T - A_I \leq -1$ , both offsets give  $\tau \leq -9/5$ ; for  $T - A_I \geq 1$ , both give  $\tau \geq 2$ . Thus a restricted violated split exists exactly when the subset-sum

instance is solvable. The rational matrix has polynomial encoding length, and restricted membership in NP is immediate. This reduction does not claim strong hardness or hardness at Boolean rank-one points.  $\square$

## 6 Gap-0 inequalities

For a rational distance vector  $d$  on  $n$  vertices, let  $Z = \mathbf{1}\mathbf{1}^\top - 2D$ , where  $D$  has off-diagonal entries  $d_{ij}$  and zero diagonal. Thus  $Z_{ii} = 1$ . A gap inequality has the form  $b^\top Zb \geq \gamma(b)^2$ , where  $\gamma(b) = \min_{s \in \{\pm 1\}^n} |s^\top b|$ .

**Lemma 6.1.** *The point  $d$  violates a gap-0 inequality if and only if  $Z$  restricted to  $s^\perp$  is not positive semidefinite for some  $s \in \{\pm 1\}^n$ . In particular, gap-0 inequalities never separate a positive semidefinite  $Z$ .*

*Proof.* An integer vector  $b$  has  $\gamma(b) = 0$  exactly when  $s^\top b = 0$  for some sign vector  $s$ . Hence a gap-0 violator gives a negative direction on  $s^\perp$ . Conversely, fix such an  $s$  and use the rational basis with columns  $e_i - s_i s_n e_n$ ,  $1 \leq i < n$ , for  $s^\perp$ . If the restricted form is not positive semidefinite, Lemma 5.1 gives a rational negative direction, which can be scaled to an integer vector  $b \in s^\perp$ . Then  $b^\top Zb < 0 = \gamma(b)^2$ .  $\square$

**Theorem 6.2.** *Deciding whether a rational distance vector violates a gap-0 inequality is NP-complete. Hardness holds even for points in the interior of the metric polytope whose matrix  $Z$  has exactly one negative eigenvalue. The reduction is from PARTITION and does not establish strong NP-hardness.*

*Proof.* Membership in NP follows from Lemma 6.1: a sign vector  $s$  is a polynomial-size certificate, and exact elimination on the displayed rational basis of  $s^\perp$  verifies that the restricted form is not positive semidefinite. It also supplies an integer violating vector of polynomial encoding length.

We reduce PARTITION [15]. Pad an instance by zeros so that it contains  $m \geq 3$  numbers, and append another  $m$  zeros. The resulting vector  $c \in \mathbb{Z}_{\geq 0}^n$  has even length  $n = 2m \geq 6$ . A partition of the original numbers can be balanced in cardinality by assigning signs to the appended zeros; conversely, a balanced partition yields a partition of the original numbers. We may restrict to  $C = \mathbf{1}^\top c > 0$ . The problem is therefore to find  $s \in \{\pm 1\}^n$  with  $s^\top c = s^\top \mathbf{1} = 0$ .

Set  $K = 4nC$ ,  $a = c + K\mathbf{1}$ ,  $A_2 = \|a\|^2$ , and write  $a_-$  and  $a_+$  for the smallest and largest coordinates of  $a$ . Define

$$\begin{aligned} \theta_* &= \frac{2(n-1)a_-^2 - (n+1)a_+^2}{(n-1)(n-2)}, & \nu &= \frac{\theta_*}{A_2 a_+^2}, \\ \rho_i &= a_i^2 \left( 1 + \frac{\nu a_i^2}{A_2} \right), & \Delta &= \sum_i \rho_i, \\ \theta_{ij} &= \frac{\rho_i + \rho_j}{n-2} - \frac{\Delta}{(n-1)(n-2)} \quad (i \neq j). \end{aligned}$$

Let  $L_\theta$  be the Laplacian with edge weights  $\theta_{ij}$ , let  $D_a = \text{diag}(a)$ , and set

$$Z = D_a^{-1} L_\theta D_a^{-1} - \nu \frac{aa^\top}{A_2}. \quad (36)$$

All numbers in this construction are rational of polynomial encoding length in  $n + \text{size}(c)$ .

First we verify the weights. Since  $a_+/a_- \leq 1 + 1/(4n)$  and, for  $n \geq 4$ ,  $(n+1)(1 + 1/(4n))^2 < 2(n-1)$ , we have  $\theta_* > 0$ . Moreover  $\theta_* \leq a_-^2/(n-1)$ , so

$$0 < \nu \leq \frac{1}{n(n-1)a_+^2} \leq \frac{1}{n}. \quad (37)$$

Writing  $\rho_-$  and  $\rho_+$  for the extreme  $\rho_i$ , we obtain

$$\theta_{ij} \geq \frac{2(n-1)\rho_- - n\rho_+}{(n-1)(n-2)} \geq \theta_* > 0,$$

because  $\rho_- \geq a_-^2$  and  $\rho_+ \leq a_+^2(1+\nu) \leq a_+^2(1+1/n)$ . A direct sum gives  $\sum_{j \neq i} \theta_{ij} = \rho_i$ . Consequently  $Z_{ii} = \rho_i/a_i^2 - \nu a_i^2/A_2 = 1$ , and  $Za = -\nu a$ .

For  $b \perp a$ , put  $h = D_a^{-1}b$ . For every real  $t$ ,

$$\sum_i a_i^2(h_i - t)^2 = \|b\|^2 + t^2 A_2 \geq \|b\|^2.$$

Since every edge weight is at least  $\theta_*$ ,

$$b^\top Zb = h^\top L_\theta h \geq n\theta_* \min_t \sum_i (h_i - t)^2 \geq \frac{n\theta_*}{a_+^2} \|b\|^2.$$

Thus  $Z$  is positive definite on  $a^\perp$  and has exactly one negative eigenvalue, namely  $-\nu$ .

If  $s^\top a = 0$ , the integer vector  $b = a$  has zero gap and  $a^\top Za = -\nu A_2 < 0$ . Conversely, suppose  $s^\top b = 0$  and  $b^\top Zb < 0$ . Write  $b = ta + b_\perp$ , with  $b_\perp \perp a$ . Since  $Za = -\nu a$ , the cross term vanishes, and

$$0 > b^\top Zb \geq -\nu t^2 A_2 + \frac{n\theta_*}{a_+^2} \|b_\perp\|^2.$$

In particular  $t \neq 0$ . Cauchy–Schwarz and  $s^\top b = 0$  give

$$|s^\top a| \leq \frac{\sqrt{n} \|b_\perp\|}{|t|} < \sqrt{\frac{\nu A_2 a_+^2}{\theta_*}} = 1.$$

The quantity  $s^\top a$  is integral, so it is zero. Finally,  $s^\top a = s^\top c + K s^\top \mathbf{1}$  vanishes exactly when both terms vanish:  $|s^\top c| \leq C < K$  prevents cancellation of a nonzero integer multiple of  $K$ . We have proved the required equivalence with balanced PARTITION.

It remains to verify the metric-polytope promise. Put  $r = a_+/a_-$ . For  $n \geq 4$ ,  $r^2 \leq 1 + 33/(64n)$  and  $r^2 < 2$ . As  $a_+ \geq K \geq 4n$ , (37) gives

$$\nu r^2 \leq \frac{1}{8n^3(n-1)} \leq \frac{1}{384n}, \quad (1+\nu)r^2 \leq 1 + \frac{199}{384n} < 1 + \frac{3}{5n}.$$

Using  $\Delta \geq na_-^2$  and  $\rho_i \leq (1+\nu)a_+^2$ , we get, for  $n \geq 6$ ,

$$\frac{\theta_{ij}}{a_i a_j} \leq \frac{2(n-1)(1+\nu)r^2 - n}{(n-1)(n-2)} < \frac{n-4/5}{(n-1)(n-2)} \leq \frac{13}{50}.$$

The last inequality is equivalent to  $(n-6)(13n-11) \geq 0$ . Also

$$\frac{\nu a_i a_j}{A_2} \leq \frac{\theta_*}{A_2^2} \leq \frac{1}{16n^4(n-1)} < \frac{1}{100}.$$



The cut coordinates from (36) are therefore

$$d_{ij} = \frac{1}{2} + \frac{\theta_{ij}}{2a_i a_j} + \frac{\nu a_i a_j}{2A_2}, \quad \frac{1}{2} < d_{ij} < \frac{127}{200}.$$

They satisfy each ordinary triangle inequality strictly, since the right-hand side exceeds 1 and the left-hand side is below 1. Every triangle perimeter is less than  $381/200 < 2$ . Thus the point lies in the interior of the metric polytope.  $\square$

The same promises hold arbitrarily close to the positive semidefinite cone. In the construction one may replace  $\nu$  by any smaller positive rational value and recompute  $\rho$  and  $\theta$ . The weight and metric bounds remain valid, and the converse estimate becomes  $|s^\top a| < \sqrt{\nu A_2 a_+^2 / \theta_*} \leq 1$ . Hence, for any prescribed positive rational  $\varepsilon$ , the reduction can enforce  $0 < \nu \leq \varepsilon$  with polynomial encoding length in the source instance and in  $\varepsilon$ . The distance to the positive semidefinite cone in spectral norm is then at most  $\varepsilon$ . Every gap-0 inequality is satisfied at positive semidefinite points.

*Remark 6.3.* If  $Z$  is not positive semidefinite, an integer negative direction gives  $b^\top Z b < 0 \leq \gamma(b)^2$ , so the decision version of unrestricted gap separation is immediately positive. This also gives a polynomial-size violating coefficient vector. Computing the exact gap of a prescribed vector is a separate partition problem; the negative-direction argument does not by itself compute the tight right-hand side of its gap inequality. At positive semidefinite points, neither the gap-0 result nor our reductions for hypermetric, gap-1 and rounded psd inequalities settle separation of all gap inequalities. The reductions do not exclude a gap violator with  $\gamma(b) \geq 3$  in a no-instance for rounded psd separation.

## 7 Consequences and limits

The main reduction identifies the obstacle to an exact separation oracle. It is not positive semidefiniteness testing, triangle separation, or recognition of a small fixed family of inequalities. Every reduction point passes those tests. Its remaining violation encodes the choice of a cover, and the proof excludes all other integer coefficient vectors. Polynomial-time enforcement of the standard relaxations therefore leaves an NP-complete separation problem.

This distinction matters when a solver adds only selected members of an infinite class. Any fixed gonality gives a polynomially enumerable family, but the exponent grows with that gonality. Theorem 3.3 proves that increasing a fixed cutoff does not make the unrestricted separation problem tractable. The rank algorithm offers a different restriction: it exploits the dimension of the rational lattice induced by the input matrix. The threshold algorithm restricts the norm of a potentially violated split normal by asking for a sufficiently large normalized violation. These are useful and distinct ways to limit a search. Neither provides a polynomial-time exact oracle at unrestricted rank and dimension.

The numerical scale of the construction is also explicit. On a yes instance, every base hypermetric or rounded positive semidefinite violation has cut-scale value  $1/(2N)$ , and every corresponding divided Boolean quadric cut or clique violation has value  $1/(4N)$ . The orthogonal extension has pure and positive odd clique violations of the same inverse-polynomial order. All coordinates have a polynomial common denominator. Hardness therefore does not depend on exponentially small slacks. The exact zero-or-positive maximum also rules out finite-factor approximation of raw violation and additive approximation at the instance-dependent accuracy in Corollary 5.10. It does not prove hardness at a fixed additive tolerance or show that every hard instance must have small violations.

Gap-0 separation concerns points outside the ellipsope: its inequalities have zero right-hand side in the  $Z$  representation and hold whenever  $Z \succeq 0$ . The remaining classification problem is full gap separation at positive semidefinite points. The present reductions control violations with gap 1 but do not exclude a violator with gap at least 3 on a no-instance. A reduction for the full family must therefore control this larger right-hand side.

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